Algebraic solutions

1 Solve each equation for the given interval. Give your answers in exact form.

a $\sin\left(x-\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$ for $-\pi \le x < \pi$

b $\cos\left(x+\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}$ for $0 \le x < 2\pi$

c $\cos\left(x-\frac{\pi}{5}\right)=\frac{\sqrt{3}}{2}$ for $0 \le x \le 3\pi$

d $\sqrt{2}\cos\left(x-\frac{\pi}{3}\right)=1$ for $-2\pi \le x \le 2\pi$

e $\tan\left(x+\frac{\pi}{5}\right)=\sqrt{3}$ for $0 \le x \le 2\pi$

f $\sin\left(3x-\frac{\pi}{3}\right)=-1$ for $-\pi \le x \le \pi$

g $2\sin\left(\frac{x}{2}+\frac{\pi}{5}\right)=1$ for $-4\pi \le x \le 4\pi$

h $\tan\left(2x-\frac{\pi}{6}\right)=-\frac{1}{\sqrt{3}}$ for $0 \le x \le \pi$

i $\tan\left(\pi x-\frac{\pi}{4}\right)=\frac{1}{\sqrt{3}}$ for $-2 \le x \le 2$

j $2\tan\left(4x+\frac{\pi}{5}\right)=1$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

k $2\sin(3x+\frac{3\pi}{2})+1=0$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

l $6\cos(3x-\frac{\pi}{2})+3=0$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

m $2\tan(2x+\frac{\pi}{9})+\frac{2}{\sqrt{3}}=0$ for $0 \le x \le \pi$

n $-2\cos(3x+\frac{\pi}{3})+1=0$ for $0 \le x \le \pi$

o $\sin^2(x-\frac{\pi}{4})-1=0$ for $0 \le x \le 2\pi$

p $12\cos^2(2x-\frac{\pi}{6})-3=0$ for $0 \le x \le \pi$

Graphical solutions

2 Consider the function $y=\cos\left(x-\frac{\pi}{6}\right)$.

a Sketch the graph of the function for $-2\pi \le x \le 2\pi$.

b Sketch the line $y=\frac{1}{2}$ on the same number plane.

c Hence, state all solutions to the equation $\cos\left(x-\frac{\pi}{6}\right)=\frac{1}{2}$ over the domain $(-2\pi, 2\pi]$. Give your answers as exact values.

3 Consider the function $y=\sin\left(x-\frac{\pi}{3}\right)$.

a Sketch the graph of the function for $-2\pi \le x \le 2\pi$..

b Sketch the line $y=\frac{1}{2}$ on the same number plane.

c Hence, state all solutions to the equation $\sin\left(x-\frac{\pi}{3}\right)=\frac{1}{2}$ over the domain $[-2\pi, 2\pi)$. Give your answers as exact values.

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4 Consider the function $y = \tan\left(x - \frac{\pi}{4}\right)$.

a Sketch the graph of the function for $-2\pi \leq x \leq 2\pi$.

b Sketch the line $y = 1$ on the same number plane.

c Hence, state all solutions to the equation $\tan\left(x - \frac{\pi}{4}\right) = 1$ over the domain $[-2\pi, 2\pi)$. Give your answers as exact values.

5 Consider the function $y = 2\sin 4x$.

a Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.

b Sketch the line $y = 1$ on the same number plane.

c Hence, state all solutions to the equation $2\sin 4x = 1$ over the domain $[-90^\circ, 90^\circ]$. Give your answers in degrees.

6 Consider the function $y = 2\sin 2x$.

a Sketch the graph of the function for $-180^\circ \leq x \leq 180^\circ$.

b State the other function you would add to the graph in order to solve the equation $2\sin 2x = 1$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $2\sin 2x = 1$ over the domain $[-180^\circ, 180^\circ]$. Give your answers in degrees.

7 Consider the function $y = 3\cos 2x + 1$.

a Sketch the graph of the function for $-\pi \leq x \leq \pi$.

b State the other function you would add to the graph in order to solve the equation $3\cos 2x + 1 = \frac{5}{2}$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $3\cos 2x + 1 = \frac{5}{2}$ over the domain $[-\pi, \pi]$. Give your answers as exact values.

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8 Consider the function $y = 2\sin 3x - 3$.

a Sketch the graph of the function for $-\frac{2\pi}{3} \leq x \leq \frac{2\pi}{3}$.

b State the other function you would add to the graph in order to solve the equation $2\sin 3x - 3 = -2$.

c Sketch the graph of this fuction on the same number plane.

d Hence, state all solutions to the equation $2\sin 3x - 3 = -2$ over the domain $\left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right]$. Give your answers as exact values.

9 Consider the function $y = -2\cos 3x$.

a Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.

b State the other function you would add to the graph in order to solve the equation $-2\cos 3x = -1$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $-2\cos 3x = -1$ over the domain $[-120^\circ, 120^\circ]$. Give your answers in degrees.

Technology

10 Solve each equation for the given interval. Round your answers to two decimal places.

a $\cos 2x = 0.9$ for $0 \leq x < 2\pi$

b $3\sin 2x = 1.2$ for $0 \leq x < \pi$

c $\tan\left(\frac{x}{2}\right) = 5$ for $-3\pi \leq x < 3\pi$

d $4\cos\left(\frac{x}{3} + 2\right) = 1$ for $0 \leq x < 4\pi$

11 Consider the equation $\sin\left(\frac{x}{2} + 60^\circ\right) = \cos\left(\frac{x}{2} - 60^\circ\right)$.

a State the two functions you would graph in order to solve this equation graphically.

b Sketch the graph of both of these functions using technology.

c Hence, state all solutions to the equation over the domain $[-360^\circ, 360^\circ]$.

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12 Use technology to solve the following functions over the interval $[0^\circ, 360^\circ)$. Give all solutions to the nearest degree.

a $2\sin 3x - 4\sin 2x = 0$

b $2\sin x + 5\cos x = 1$

c $\cos 3x + \cos x = \sin x$

d $2\cos 3x - 3\cos 2x = 0$

e $\cos\left(\frac{x}{2}\right) - 4\sin 2x = 0$

f $\sin\left(\frac{x}{2}\right) = 7\cos 3x$

g $4\sin^5(x) = -(\cos x + 1)$

h $\cos^3(x) + \cos x = -1$

i $\sin 3x = \ln 3x$

j $e^x - 2 = \cos x$

13 Use technology to solve the following equations for $0 \leq x \leq 2\pi$. Give all solutions to three decimal places.

a $2\sin x + 5\cos x = 1$

b $3\sin 2x + 2\sin 3x = 0$

c $3\sin(2x) = x + 1$

d $2\cos 3x + 3\cos 2x = 0$

e $\cos 3x + \cos x = \sin x$

f $\cos\left(\frac{x}{2}\right) - 4\sin 2x = 0$

g $\sin\left(\frac{x}{2}\right) = 5\cos 3x$

h $2\tan(x - \frac{\pi}{3}) = -\frac{x}{2}$

i $4\sin^5(x) = -(\cos x + 1)$

j $\cos^5(x) + \cos x = -1$

k $e^x - 6 = \cos x$

l $\sin 2x = \ln 2x$

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